

Tutorial Quiz 2 Rubric

Q1] Find the gradient of the function

$$f(x, y) = y - x$$

at point $(2, 1)$

The gradient of a function $f(x, y)$ is :

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$$

for us

$$\frac{\partial f}{\partial x} = -1 \quad ; \quad \frac{\partial f}{\partial y} = 1$$

$$\therefore \nabla f = (-1, 1) \quad \text{--- (1)}$$

Q2 Find the local maxima, minima & saddle point of the f^n

$$g(x, y) = x^2 + 2xy$$

First we find the first partial derivatives:

$$\frac{\partial f}{\partial x} = 2x + 2y \quad ; \quad \frac{\partial f}{\partial y} = 2x$$

Then we equate to zero to find the critical points!

$$\frac{\partial f}{\partial y} = 0 = 2x \Rightarrow x = 0$$

$$\text{substituting in } \frac{\partial f}{\partial x} = 2(0) + 2y = 0$$

$$\therefore \Rightarrow y = 0$$

$\therefore (0, 0)$ is a critical point. — (1)
[1 mark for]
critical point

Now we compute the second partial derivatives

$$\frac{\partial^2 f}{\partial x^2} = 2 \quad ; \quad \frac{\partial^2 f}{\partial y^2} = 0$$

$$\frac{\partial^2 f}{\partial x \partial y} = 2$$

substituting in & computing the discriminant

$$D = \frac{\partial^2 f}{\partial x^2} \cdot \frac{\partial^2 f}{\partial y^2} - \left[\frac{\partial^2 f}{\partial x \partial y} \right]^2$$

$$= 2 \cdot (0) - (2)^2$$

$$= -4 < 0$$

Since $D < 0$ there is only a saddle point, no minima / maxima.

Saddle at $(0, 0)$ ————— (1)

[either statements]
get 1 mark